Assignment - 3

(Q) The potential on the surface of a hollow sphere is given as : (radius = R)

$$V_0(\theta) = k \sin^2\left(\frac{\theta}{2}\right)$$

Find the potential inside the hollow sphere

Ans: Let $V(r,\theta)$ be the potential inside the sphere

$$\nabla^2 V = 0$$

In spherical coordinates :

$$\nabla^2 V = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial V}{\partial r}\right) + \frac{1}{r^2 \sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta \frac{\partial V}{\partial \theta}\right) + \frac{1}{r^2 \sin^2\theta}\frac{\partial^2 V}{\partial \phi^2}$$

$\because$ V does not depend on ϕ ; $\frac{\partial V}{\partial \phi} = 0 \Rightarrow \frac{\partial^2 V}{\partial \phi^2} = 0$

Let $V(r,\theta) = R(r) \cdot \alpha(\theta)$

$$\therefore \nabla^2 V = \frac{\alpha}{r^2}\frac{d}{dr}\left(r^2 \frac{dR}{dr}\right) + \frac{R}{r^2 \sin\theta}\frac{d}{d\theta}\left(\sin\theta \frac{d\alpha}{d\theta}\right) = 0$$

$$\Rightarrow \frac{1}{R}\frac{d}{dr}\left(r^2 \frac{dR}{dr}\right) + \frac{1}{\alpha \sin\theta}\frac{d}{d\theta}\left(\sin\theta \frac{d\alpha}{d\theta}\right) = 0$$

[multiplying by $\frac{r^2}{R\alpha}$]

$$\Rightarrow \frac{1}{R}\frac{d}{dr}\left(r^2 \frac{dR}{dr}\right) = -\frac{1}{\alpha \sin\theta}\frac{d}{d\theta}\left(\sin\theta \frac{d\alpha}{d\theta}\right)$$

$$= l(l+1)$$

$$\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\alpha}{d\theta}\right) + l(l+1)\alpha = 0 \quad \text{— (a)}$$

$\boxed{\text{RHS}}$:

Let $x = \cos\theta$; $\alpha(\theta) = y(x)$.

$$\Rightarrow \boxed{\frac{dx}{d\theta} = -\sin\theta} \text{ — (i)} \qquad \Rightarrow \frac{d\alpha}{d\theta} = y' \cdot \frac{dx}{d\theta} = -\sin\theta\, y' \text{ — (ii)}$$

$$\Rightarrow \frac{d^2\alpha}{d\theta^2} = -\cos\theta\, y' - \sin\theta\, y'' \cdot \frac{dx}{d\theta} \quad [\text{using (i)}]$$

$$= -\cos\theta\, y' + \sin^2\theta\, y''$$

$$\Rightarrow \boxed{\frac{d^2\alpha}{d\theta^2} = -xy' + (1-x^2)y''} \text{ — (iii)}$$

$$\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\alpha}{d\theta}\right) = \frac{1}{\sin\theta}\left[\sin\theta\frac{d^2\alpha}{d\theta^2} + \cos\theta\frac{d\alpha}{d\theta}\right]$$

$$= \frac{1}{\sin\theta}\left[\sin\theta\left(-xy' + (1-x^2)y''\right) + \sin\theta\cos\theta\, y'\right]$$

[using (i), (ii) & (iii)]

$$= -xy' + (1-x^2)y'' - \cos\theta\, y'$$

$$= (1-x^2)y'' - 2xy'$$

$$\therefore (1-x^2)y'' - 2xy' + l(l+1)y = 0.$$

This is the Legendre ODE. Its solutions are the Legendre Polynomials

$$y(x) = P_l(x) \Rightarrow \boxed{\alpha(\theta) = P_l(\cos\theta)}$$

$\boxed{\text{LHS}}$:

$$\frac{1}{R}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) - l(l+1)R = 0.$$

$$\Rightarrow r^2R'' + 2rR' - l(l+1)R = 0 \text{ — (b)}$$

Let $R = r^s \Rightarrow R' = s r^{s-1}$

$$\Rightarrow R'' = s(s-1)r^{s-2}$$

Substituting into (b):

$$s(s-1)r^s + 2sr^s - l(l+1)r^s = 0.$$

$$\Rightarrow s(s+1) - l(l+1) = 0 \quad [r^s \neq 0]$$

$$\Rightarrow (s-l)(s+l+1) = 0$$

$$\therefore s = l \quad \text{or} \quad s = -(l+1).$$

$$\boxed{R(r) = A_l r^l + B_l r^{-(l+1)}}$$

As $r \to 0$; ~~$R(r) \to \infty$~~ $r^{-(l+1)} \to \infty$

This is not feasible inside the sphere. Hence, $\boxed{B_l = 0}$

$\therefore$ $\boxed{R(r) = A_l r^l}$

$$\therefore \quad V(r,\theta) = \sum_{l=0}^{\infty} A_l r^l P_l(\cos\theta)$$

@ $r = R$: $V_0(R,\theta) = k \sin^2\left(\frac{\theta}{2}\right)$

$$= \frac{k}{2}(1 - \cos\theta)$$

It only contains terms of $P_0(\cos\theta) = 1$ & $P_1(\cos\theta) = \cos\theta$

Hence only $l=0$ & $l=1$ terms appears in $V(r,\theta)$,

$\therefore$ ~~$V(r,\theta)$~~ as all the Legendre polynomials are orthogonal to each other.

$$\therefore \quad V(r,\theta) = A_0 + A_1 r P_1(\cos\theta)$$

$$= A_0 + A_1 r \cos\theta$$

$$\therefore \quad V_0(R,\theta) = A_0 + A_1 R \cos\theta = \frac{k}{2} - \frac{k}{2}\cos\theta$$

Comparing both sides:

$$A_0 = \frac{k}{2} \quad \& \quad A_1 = \frac{-k}{2R}$$

$$\therefore \quad \boxed{V(r,\theta) = \frac{k}{2} - \frac{kr}{2R}\cos\theta} \quad (\text{Ans})$$